General-Relativity

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Contents

Lecture 1: Syllabus day

Tue 02 Sep 2025 12:31

We will be using Carroll. Course webpage at `https://physics.bu.edu/~fitzpatr/f25`. Grad class so please call him Liam. Most of the first half of the course is developing the theory of differential geometry, in order to discuss the manifold that is spacetime. Weinberg's book on GR is a good resource as well. Gravity is described by a (not quite) Gauge field $h_{\mu\nu}$ which we will discuss at the end of the course. This implies the equivalence prinicpal which implies gravity behaves as if spacetime is a Riemannian manifold, but it is more fundamental meaning the geometry is not the truth of the universe in the physical sense, only the mathematical sense.

Newtonian gravity is not Lorentz invariant. Fixing this requires the equivalence principle, which states the laws of physics are the same in any inertial reference frame. A great way to phrase this is that standing in a box accelerating upwards is the same as standing in a box that is on Earth, subject to a downwards pull. This means gravity may be described by outwards acceleration.

Interpret gravity as outwards acceleration in this way. This then amounts to sowing together *different* accelerating frames around the border of the Earth. Wow I wonder if this is related to manifolds?

Chapter 1

Review of special relativity

We take the convention that x^μ is a 4-vector with $x^0 = t$, $x^1 = x$, $x^2 = y$, and $x^3 = z$.

Spacetime is obviously a 4-manifold, but it admits a metric, since distance can be measured. The speed of light is the same in all reference frames, so if we fix two coordinate systems x^μ and $(x')^\mu$ for spacetime, they must agree on the speed of light. More precisely, given an initial point x_0^μ and final point x_f^μ for a ray of light, if $\Delta x^0 = x_f^0 - x_0^0$ is the distance in x (and the same for other coordinates), then the time elapsed must be

$$c\Delta t = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}.$$

Equivalently,

$$0 = c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2.$$

Invoking that the speed of light is the same in these coordinate systems,

$$0 = c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2 = c^2\Delta(t')^2 - \Delta(x')^2 - \Delta(y')^2 - \Delta(z')^2.$$

We define this quantity as the invariant distance between x^μ and $(x')^\mu$. Equivalently,

$$\Delta s^2 = \eta_{\mu\nu}\Delta x^\mu \Delta x^\nu,$$

where

$$\eta_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1)$$

is the Minkowski metric. Here we recall Einstein notation that repeated indicies indicate implicit summations. Since $\eta_{\mu\nu} \neq 0$ iff $\mu = \nu$, the claimed equality follows.

Remark 1.0.1. Carroll uses $x^{\mu'}$ instead of $(x')^\mu$ for different coordinate symbols. This is just stupid because μ' should still be μ????? Like why would we ever do this wtf

Definition 1.0.2. The manifold $\mathbb{R}^4$ with topology given by Δs^2 is *Minkowski space*. The spacetime metric is also known as the *Minkowski metric*. I will often denote this space by $\mathcal{M}$ for simplicity; this is nonstandard as far as I am aware, but I like naming my mathematical objects.

(Minkowski is pronounced "Minkovski"). When we refer to "flat space" in this course, we always mean Minkowski space. As we do in geometry, we now consider symmetries of flat space. Clearly translations $x^\mu \mapsto (x')^\mu = x^\mu + a^\mu$ for any fixed $a^\mu \in \mathcal{M}$ are metric-invariant homeomorphisms of Minkowski space. However, we are also interested in the Lorentz tranformations

Definition 1.0.3. Lorentz transformations Λ^μ_ν are linear maps on x^μ that preserve Δs^2. They are given by $x^\mu \mapsto \Lambda^\mu_\nu x^\nu$, where the tensor Λ^μ_ν is a 4×4 matrix we describe later.

We equivalently have

$$\Delta x^\mu \Delta x^\nu \eta_{\mu\nu} = \Lambda^\mu_\alpha \Delta x^\alpha \Lambda^\nu_\beta \Delta x^\beta \eta_{\mu\nu}.$$

We thus need

$$\eta_{\alpha\nu} = \Lambda^\mu_\alpha \Lambda^\nu_\beta \eta_{\mu\nu}.$$

Lorents transformations are any map such that this is true.

Definition 1.0.4. We don't necessarily need to fix Δs^2, we need to lead the metric for *light rays* invariant, where $\Delta s^2 = 0$. Maps that do this are called *conformal transformations*.

We are using a larger set than usual for Lorentz transformations. Oftentimes linear transformations must be homotopic to the identity, but our definition is more general. *Discrete symmetries* do this. For example, *time parity* $T : x^0 = t \mapsto -t$. Of course T is not homotopic to 1, but it is clearly a metric-invariant homeomorphism. In general, any spacial parity $x^i \mapsto -x^i$ with $i \in \{1, 2, 3\}$ fixed is also a transformation.

Oftentimes we do not want to discuss discrete transormations, so we impose the following requirements:

1. $\Lambda^0_0 \geq 1$,
2. $\det \Lambda = 1$.

The first gets rid of $t \mapsto -t$, and the second kind of just refuses stretching.

Notation 1.0.5. We will often have to deal with more general metrics than just Minkowski's metric. We will generally use the notation $g_{\mu\nu}$, and we define $g^{\mu\nu} \mapsto g^{-1}_{\mu\nu}$, meaning

$$g^{\mu\nu} g_{\nu\rho} = \delta^\mu_\rho = 1,$$

where δ^μ_ρ is the Kronecker delta $\delta^\nu_\mu = 1$ iff $\mu\nu$, 0 otherwise. Everything else raises and lowers indices with the metric, meaning

$$x^\mu := g^{\mu\nu} x_\nu, \qquad x_\mu := g^{\mu\nu} x^\nu.$$

The Minkowski metric is self-inverse $\eta_{\mu\nu} = \eta^{\mu\nu}$. Using this notation, viewing $\eta_{\mu\nu}$ as a vector in μ, we can define

$$\eta^\mu_\nu := \eta^{\mu\alpha} \eta_{\alpha\nu} = 1.$$

For this reason, we will almost never write g^ν_μ or g^μ_ν, since they are both 1.

Lecture 2: Vectors and dual vectors

Thu 04 Sep 2025 12:31

In order to define vectors on a manifold, we work with the *tangent space*. Let M be an n-manifold, and choose some path $x^\mu(-)$ parametrized by the variable λ. Composing along the charts gives some $g : M \to \mathbb{R}$, and we can compute partial derivatives

$$\frac{\partial g(x)}{\partial x^\mu}$$

by once again composing along the charts in the other direction, which yields some $\mathbb{R}^n \to \mathbb{R}$. The atlas condition guarentees invariance under choice of charts. The tangent space at $p = x^\mu(\lambda)$, written as T_pM, is the $\mathbb{R}$-vector space of these derivatives. It is spanned by

$$\left\{ \frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n} \right\}.$$

Any vector v of this space admits a basis decomposition

$$v = v^\mu \frac{\partial}{\partial x^\mu}.$$

There is a vector

$$\frac{\partial}{\partial x^\mu},$$

with νth component

$$\left(\frac{\partial}{\partial x^\mu} \right)^\nu = \delta^\nu_\mu \left(\frac{\partial}{\partial x^\mu} \right).$$

Then

$$\frac{\partial}{\partial x^\mu} = \left(\frac{\partial}{\partial x^\nu} \right)^\mu \frac{\partial}{\partial x^\mu}.$$

Oftentimes we write

$$\hat{e}_\mu = \frac{\partial}{\partial x^\mu} = \hat{e}^\nu_\mu \frac{\partial}{\partial x^\nu}$$

for simplicity.

Given two coordinate systems $x^\mu, (x')^\mu$, the tangent space T_pM takes both $\frac{\partial}{\partial x^\mu}$ and $\frac{\partial}{\partial (x')^\mu}$ as bases. To translate between bases, we use t he chain rule

$$\frac{\partial}{\partial x^\mu} = \frac{\partial (x')^\alpha}{\partial x^\mu} \frac{\partial}{\partial (x')^\alpha}.$$

We thus obtain a change of basis: given

$$v = v^\mu \frac{\partial}{\partial x^\mu} = (v')^\mu \frac{\partial}{\partial (x')^\mu},$$

we have

$$v = v^\mu \frac{\partial (x')^\alpha}{\partial x^\mu} \frac{\partial}{\partial (x')^\alpha} = (v')^\alpha \frac{\partial}{\partial (x')^\alpha}.$$

Thus,

$$v^\mu \frac{\partial (x')^\alpha}{\partial x^\mu} = (v')^\alpha.$$

We often have functions of vectors. We write $f(v)$ by $f(v^\mu)$, and leave the basis component implicit (i.e. not writing $f(v^1, \dots, v^n)$).

Let $x^\mu(\lambda)$ be a path in M. We define the *total* derivative

$$\frac{\mathrm{d}x^\mu(\lambda)}{\mathrm{d}\lambda}$$

with components given by the components in the tangent space of $\frac{\mathrm{d}x}{\mathrm{d}\lambda}$. The total derivative can indeed be computed since paths are functions in 1 variable. Viewed in the tangent space, given

$$V \coloneqq \frac{\mathrm{d}x}{\mathrm{d}\lambda},$$

we have

$$V = V^\mu \frac{\partial}{\partial x^\mu}.$$

How does V act on a function g on the manifold?

$$V g(x) = V^\mu \frac{\partial}{\partial x^\mu} g(x) = \frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda} \frac{\partial}{\partial x^\mu} g(x) = \frac{\mathrm{d}}{\mathrm{d}\lambda} g(x(\lambda))$$

where the last equality is precisely the reverse of the chain rule.

We have a metric $g_{\mu\nu}$. Given vectors v^μ, w^ν, we can compute the *inner product* $\langle v, w\rangle = g_{\mu\nu} v^\mu w^\nu$, and we write $g(v, w)$ for this inner product. We generally write this as

$$v^\mu w_\mu, \qquad w_\mu \coloneqq g_{\mu\nu} w^\nu.$$

This agrees with our previous raising-lowering indices convention. We call vectors w_μ *dual vectors*. Dual vectors have a *dual basis* $\mathrm{d}x^\mu$ given by

$$\mathrm{d}x^\mu \left(\frac{\partial}{\partial x^\nu} \right) = \delta^\mu_\nu.$$

Dual vectors essentially act on vectors, rather than on functions. A general dual vector ω admits a basis decomposition via

$$\omega = \omega_\mu \mathrm{d}x^\mu.$$

Linearity thus extends the formula for dual vectors acting on vectors:

$$\omega(v) = \omega_\mu \mathrm{d}x^\mu \left(v^\nu \frac{\partial}{\partial x^\nu} \right) = x_\mu v^\nu \mathrm{d}x^\mu \frac{\partial}{\partial x^\nu} = \omega_\mu v^\nu \delta^\nu_\mu = \omega_\mu v^\mu.$$

Consider transformations under basis change. Choose $x^\mu, (x')^\mu$ coordinates. Since $\omega(v) = \omega_\mu v^\mu$ is basis independent,

$$\omega_\mu v^\mu = \omega'_\alpha (v')^\alpha \implies \omega'_\alpha = \omega_\nu \frac{\partial x^\nu}{\partial (x')^\mu} \implies \omega'_\alpha (v')^\alpha = \omega_\nu \frac{\partial x^\nu}{\partial (x')^\alpha} \left(v^\beta \frac{\partial (x')^\alpha}{\partial x^\beta} \right)$$

$$= v_\nu \delta^\nu_\beta v^\beta = \omega_\nu v^\nu.$$

We finally are able to rigorize differentials. Let $\varphi(x)$ a function of M, and recall we define

$$\mathrm{d}\varphi = \frac{\partial \varphi(x)}{\partial x^\mu} \mathrm{d}x^\mu.$$

Some books write

$$\varphi, \mu = \frac{\partial \varphi}{\partial x^\mu}.$$

Question: are "dual vectors" actually elements (at least, up to isomorphism) of the dual vector space?

Lecture 3: Locally inertial coordinates

Tue 09 Sep 2025 12:29

We will now begin discussing tensors. We begin with the metric tensor $g_{\mu\nu}$. Recall that components of the dual vector transform by

$$\omega_\mu \to \omega'_\mu = \frac{\partial x^\alpha}{\partial (x')^\mu}\omega_\alpha.$$

The metric tensor transforms the same way in 2 indices, meaning

$$g_{\mu\nu} \to g'_{\mu\nu} = \frac{\partial x^\alpha}{\partial (x')^\mu}\frac{\partial x^\beta}{\partial (x')^\nu}g_{\alpha\beta}.$$

This implies there is some basis independent quantity g, and the components $g_{\mu\nu}$ are a choice in a basis, meaning

$$g = g_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu = g'_{\mu\nu}\mathrm{d}(x')^\mu \mathrm{d}(x')^\nu.$$

This means the metric's components are not invariant under coordinate change. This is useful: let M be spacetime, and choose a point x_p^μ in M. WLOG, there exist coordinates such that $g_{\mu\nu}(x_p^\mu) = \eta_{\mu\nu}$ and such that the partials are 0:

$$\partial_p g_{\mu\nu}(x_p^\mu) = 0,$$

where we have introduced the very convenient notation

$$\partial_\mu = \frac{\partial}{\partial x^\mu}.$$

We call such coordinates *locally inertial coordinates.*

Let x^α and $(x')^\mu$ be coordinate systems. Fix x_p^μ a point, and compute the Taylor expansion $(x')^\mu$ about x_p^μ:

$$(x')^\mu(x) = (x')^\mu(x_p) + (x-x_p)^\alpha(\partial_\alpha (x')^\mu(x_p)) + \frac{1}{2}(x-x_p)^\alpha(x-x_p)^\beta(\partial_\alpha\partial_\beta (x')^\mu_p) + \cdots .$$

The first few terms give us the fact that $g_{\mu\nu}$ has $\frac{d(d+1)}{2}$ parameters, but we may choose d^2 total free parameters. Thus, we may choose $g_{\mu\nu} = \eta_{\mu\nu}$ and still have $\frac{d(d-1)}{2}$ free parameters left over. Physically, these free parameters that leave the Minkowski metric invariant are the Lorentz transformations: velocity boosts and rotations. We think of these Lorentz transformations as "generating" elements: a rotation in $\mathbb{R}^2$ for example should be a "generating element" (interpreted as an infintesimal) for SO(2).

We will almost always assume locally inertial coordinates. We will choose a point p, and assume WLOG that the first derivatives vanish at that point and the metric at that point is Minkowski. This is useful because we know the laws of physics in Minkowski space.

Chapter 2

Tensors

We will now introduce tensors in a differential geometric context. We have already seen one: $g = g_{\mu\nu} \mathrm{d}x^\mu \otimes \mathrm{d}x^\nu$. Earlier, we didn't write the tensor product, but now we do. This is an element of the tensor square of the dual space $(T_p M^\vee)^{\otimes 2}$. Similarly, $g^{-1} = g^{\mu\nu} \partial_\mu \otimes \partial_\nu$ is an element of the tensor product of the tangent space. We may also write $1 = \delta^\nu_\mu \partial_\mu \otimes \mathrm{d}x^\nu$. These are called $(0,2)$-, $(2,0)$-, and $(1,1)$-tensors, respectively. We will never again write the tensor product and will leave it implicit.

Of course, we may generalize this to (r,s)-tensors which are given as

$$T = T^{\mu_1 \ldots \mu_r}_{\nu_1 \ldots \nu_s} \partial_{\mu_1} \cdots \partial_{\mu_2} \mathrm{d}x^{\nu_1} \cdots \mathrm{d}x^{\nu_s}.$$

These of course may be interpreted as functions on the relevant space. Matrices may be viewed as bilinear via the identification $M(v_1, v_2) \coloneqq M_{ij} v_1^i v_2^j$. (r,s)-tensors generalize this in the sense that they are (r,s)-linear functions: they are r-multilinear in vectors and s-multilinear in dual vectors by acting on vectors in precisely the same way. We can always use the metric to raise and lower indices, meaning we can obtain a $(0, r+s)$-tensor (or likewise in the tangent space tensor power) by lowering indices on an (r,s)-tensor:

$$T_{\mu_1 \ldots \mu_r \nu_1 \ldots \nu_s} = g_{\mu_1 \alpha_1} \ldots g_{\mu_r \alpha_r} T^{\alpha_1 \ldots \alpha_r}_{\nu_1 \ldots \nu_r}.$$

We may also be interested in making a tensor T into a symmetric tensor, wherein $T_{ij} = T_{ji}$ (without writing extra indices), or antisymmetrizing it by $T_{ij} = -T_{ji}$. We can do this with matrices: given M_{ij} a matrix,

$$\frac{M_{ij} + M_{ji}}{2}$$

is symmetric, and subtracting gives an antisymmetric matrix. We do a similar process with matrices: given σ any permutation on $\{1, \ldots, n\}$, we write

$$T_{(\mu_1 \ldots \mu_n)} \coloneqq \frac{1}{n!} \sum_\sigma T_\sigma.$$

We only symmetrize on things in the parenthases, meaning

$$T_{(\mu\nu)\rho} = \frac{1}{2} \left(T_{\mu\nu\rho} + T_{\nu\mu\rho} \right)$$

only symmetrizes μ, ν. Similarly, $T_{(\mu|\nu|\rho)}$ means we symmetrize μ and ρ; the line is a "break". Notation for antisymmetrizing is given by brackets

$$T_{[\mu\nu]} = \frac{1}{2}(T_{\mu\nu} - T_{\nu\mu}).$$

These are important due to their relation to Lorentz invariance. For example, we apply this to Maxwell's equations. We write these componentwise as

$$\varepsilon^{ijk}\partial_j B_k - \partial_0 E^i = J_E^i,$$

$$\partial_i E^i - J_E^0,$$

$$\varepsilon^{ijk}\partial_j E_k + \partial_0 B^i = J_M^i.$$

$$\partial_i B^i = -J_m^0.$$

Lecture 4: idk

Thu 11 Sep 2025 12:36

Missed some stuff, read notes.

The energy-momentum tensor or stress tensor $T_{\mu\nu}$ is how energy and momentum enter into the equations of motion, since the stress tensor turns energy and momentum and turns them into an object which transforms like a tensor.

A simple example is the case of a fluid. We can view this as a collection of individual molecules, and (in theory) we could keep track of the trajectory of each particle as some $x^\mu(\lambda) = (t(\lambda), x^i(\lambda))$. There are many ways to parameterize x^μ, so we choose to parameterize it with the *proper time* τ. This is the amount of time that elapses for the *particle itself*: if it was carrying a stopwatch, it's the readout of the stopwatch. Using Δs, we find

$$\left(\frac{\mathrm{d}\tau}{\mathrm{d}\lambda}\right)^2 = -\eta_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}.$$

This basically puts the particle in its rest frame. In the rest frame,

$$\frac{\mathrm{d}x^i}{\mathrm{d}\lambda} = 0$$

since the particle is not moving. Therefore,

$$0 = -\eta_{\mu\nu}\mathrm{d}x^\mu\mathrm{d}x^\nu = -\eta_{00}\mathrm{d}x^0\mathrm{d}x^0 = \mathrm{d}t^2.$$

Convention: particle physicists take $(+,-,-,-)$ for $\eta_{\mu\nu}$, and relativists take the opposite convention $(-,+,+,+)$. We are going to swap to the relativistic conventions now. This is why there is a $-\eta_{\mu\nu}$ in the equation.

Therefore, in the rest frame,

$$\mathrm{d}\tau^2 = \mathrm{d}t^2 \implies \left(\frac{\mathrm{d}\tau}{\mathrm{d}\lambda}\right)^2 = \left(\frac{\mathrm{d}t}{\mathrm{d}\lambda}\right)^2.$$

We can also view fluids macroscopically. The quantity

$$u^\mu := \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}$$

can be viewed as a continuous function of spacetime by viewing it as an average over the little x^μs. We then have

$$u^\mu u_\mu = \eta_{\mu\nu}\eta^\mu\eta^\nu = \eta_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} = \eta_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}\tau}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\frac{\mathrm{d}\tau}{\mathrm{d}\lambda}.$$

Recall of course that in the rest frame, the space components go to 0 and τ is proper time, so plugging in our previous equations we find it normalizes to -1.

Let's find $T_{\mu\nu}$ for a perfect fluid. The stress tensor locally depends on u^μ, and in the local rest frame its rotationally invariant. We first look at $T^{\mu\nu}$ in the rest frame. This is a symmetric 4x4 matrix. For it to be rotationally invariant, all non-diagonal elements have to vanish. This can be proven by some annoying math. We also find that due to $\eta_{\mu\nu}$ signs, the first term $\rho = T^{00}$ is energy density, and the other terms T^{ii} are pressure. This follows because space and time are different:

In a perfect fluid, x, y, z are all the same, so rotations in 3d don't do anything, and only 1 is invariant under 3d rotations. We are not demanding that we do 4-dimensional rotations (boosts), since it is not invariant under these. This gives us a second degree of freedom in ρ.

Aside: If we required it be invariant under boosts, we would have $\rho = -p$ due to the metric. This would imply that for a Lorentz transformation Λ,

$$T^{\mu\nu} = \Lambda\eta^{\mu\nu} = \begin{pmatrix} -\Lambda & & & \\ & \Lambda & & \\ & & \Lambda & \\ & & & \Lambda \end{pmatrix}.$$

This is dark energy apparently.

We have simplified $T^{\mu\nu}$ to

$$T^{\mu\nu} = \mathrm{diag}(\rho, p, p, p).$$

We then lower indices

$$T_{\mu\nu} = \eta_{\mu\alpha}\eta_{\nu\beta}T^{\alpha\beta}.$$

Useful trick. We start by deriving a formula in a specific frame, and we then use tensor analysis to write the tensor in any frame. We will now do this with $T_{\mu\nu}$, since we know it in its rest frame. We start by writing $T^{\mu\nu}$ in terms of $u^\mu = (1, 0, 0, 0)$ in the rest frame. We of course have $u^\mu u^\nu = \mathrm{diag}(1, 0, 0, 0)$, so we can start with $\rho u^\mu u^\nu$.

The trick requires us to only use the 2-tensor $\mu\nu$, so writing it down we have to use thee two indices only. Thus, the things we can put on the right-hand side are $u^\mu u^\nu$ and $\eta^{\mu\nu}$. In other words, the fluid picked out a specific reference frame, so we can only use things which pick out the same reference frame, which are u and the spacetime metric, since the latter is invariant under reference frames. We have

$$p\eta^{\mu\nu} + (\rho + p)u^\mu u^\nu = T^{\mu\nu}.$$

This gives us a way to express $T^{\mu\nu}$ in a general frame, by invariance under coordinate transformations. We simply have

$$T^{\mu\nu} = p\eta^{\mu\nu} + (\rho + p)u^\mu u^\nu$$

in *any* frame. Finally, note that energy and momentum are conserved, meaning we must have

$$\partial_\mu T^{\mu\nu} = 0.$$

This follows for this example without extra work I think.

Chapter 3

Equivalence principle

Our previous discussion forms the basis for the equivalence principle, which is so important that it is basically a definition.

Definition 3.0.1. The *weak equivalence principle* states that forces on particles only depend on distance:

$$m\mathbf{a}_i = \mathbf{F}_i = -m\Delta\Phi_i = \sum_{j\neq i}\mathbf{F}(\mathbf{x}_i - \mathbf{x}_j).$$

We can thus always remove gravity by passing to an accelerating frame: given the old coordinate system x^μ, we map to a new system $(x')^\mu$ by

$$x^\mu \to x^\mu + \frac{1}{2}t^2\Delta\Phi.$$

The *strong* equivalence principle states that all laws of physics are exactly the same in any (inertial) reference frame, and are invariant under coordinate transformations.

Mass comes from interactions. The mass of the proton, for example, is 1GeV, but the mass of quarks are only a few MeV. The mass of the proton comes largely from the interactions of its quarks, so forces have to couple to these interactions such that the Newtonian weak equivalence principle is satisfied. Thus, mass is inherentely relativistic, and it's hard for the strong equivalence principle to not be true.

For example, effects of gravity are precisely effects of non-gravitational acceleration. This allows us to convert problems of gravity to problems that we already know: acceleration without gravity. Let's say one twin is standing on Earth and one twin is *accelerating* in a rocket. The twin that is standing still is normal, but the moving twin experiences time dilation. If both are in inertial frames, nothing weird happens, but if the other accelerates to come back, the one who was moving is younger. This tells us that if we switch this scenario to being in a gravitational field, we find that we age slower in gravitational fields.

Lecture 5: Covariance

Tue 16 Sep 2025 12:32

Before discussing covariance, we continue our discussion of the Equivalence principle. Consider a laser firing a photon at a fixed wavelength λ_i. We measure it at the top with a detector to be a *different number* $\lambda_f \neq \lambda_i$. This experiment was done at Harvard by shooting photons 22 meters. This small difference in gravity was able to measure the redshift. We derive this using only the equivalence principle.

This situation is equivalent to the entire tower accelerating upwards at $9.8m/s^2$, and shooting a laser up the tower the instant it starts accelerating. The distance $z(t)$ of the photon from z_0 is $z(t) = z_0 + \frac{1}{2}at^2$. Of course, $\Delta t = t_f - t_i = \frac{z_f}{c}$. We choose $t_i = 0$. The velocity of the rocket is simply $v = at_f = a\frac{z_0}{c}$. There is then a redshift factor of $\frac{\Delta\lambda}{\lambda_i}$ by the Doppler effect. This gives a doppler shift of

$$\sqrt{\frac{1+v/c}{1-v/c}} - 1.$$

This approximates out to az_f/c^2 in the small gravitational potential when plugging in the ts.

We use a mathematical systemization of this argument, called *general covariance*.

1. First, write down equations of motion for physics in the absense of gravity.
2. Then, introduce the metric and its derivatives into the equations of motion in such a way that the resultant equations of motion are invariant under coordinate changes (Lorentz transformations). We then WLOG reduce to Minkowski space.

To do this, we need to learn how to efficiently write the equations of motion in a coordinate-invariant manner more efficiently. We will explore this at length. The fascinating implications of this fact are that gravity follows completely by coordinate-invariance and the Minkowski metric, meaning the *entire effect of gravity* is the coordinate transformation into the gravitational frame.

We start by considering free motion of a particle

$$\frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} = 0.$$

Recall that u^μ is the 4-velocity (u^0, u^i) and τ is the proper time (rest frame time)

$$\mathrm{d}\tau^2 = -\eta_{\mu\nu}\mathrm{d}x^\mu\mathrm{d}x^\nu.$$

. Idk τ represents time in our coordinate system. This is equivalently written as

$$\frac{\mathrm{d}^2x^\mu}{\mathrm{d}\tau^2} = \left(\frac{\mathrm{d}^2t}{\mathrm{d}\tau^2}, \frac{\mathrm{d}^2x^i}{\mathrm{d}\tau^2}\right) = 0.$$

In the rest frame, $t = \tau$, so the time component vanishes in the rest frame.

The issue is we want to also work with massless particles, and they don't have proper time. This means we can't use τ since it never changes, so we have to do something else. Define $\sigma = x^0$ as the time component in an inertial frame that is not necessarily the reference frame. We then have

$$\frac{\mathrm{d}^2x^\mu}{\mathrm{d}\sigma^2} = 0.$$

We use affine parameters to show that σ is invariant under our choice. Write $\sigma = a\lambda + b$, where λ is an affine parameter. We then still have

$$\frac{\mathrm{d}^2x^\mu}{\mathrm{d}\lambda^2} = 0$$

by change of variables. We thus have the freedom to redefine the coordinate system using any affine parameter.

Both massive *and* massless particles can then be written as

$$\frac{\mathrm{d}^2\xi^\mu}{\mathrm{d}\lambda^2} = 0, \qquad \xi^\mu = \text{freely falling frame.}$$

Of course by Lorentz invariance (and in particular, invariance under boosts), we see that the velocity of the particle in this frame need not be defined, since all are Lorentz invariant.

The question is: what are the equations of motion in the lab frame? The answer is given by transforming

$$\frac{\mathrm{d}^2\xi^\mu}{\mathrm{d}\lambda^2} = 0 \to \text{lab frame equivalent.}$$

We do this explicitly by computing $\xi^\mu(x)$. We want to write these in terms of the metric. In the freely falling frame we choose the metric by $\eta_{\mu\nu}$ with vanishing first derivatives. This is what the freely falling frame is (recall it exists by earlier discussion). The lab frame is an accelerating frame because our lab is on a rocketship. I love GR.

Write $g_{\mu\nu}$ for the metric for the lab frame, and recall that we have that our metric transformation is given by

$$g_{\mu\nu} \to g'_{\mu\nu} = g_{\alpha\beta}\frac{\partial x^\alpha}{\partial(x')^\mu}\frac{\partial x^\beta}{\partial(x')^\nu} = \partial_\mu x^\alpha \partial_\nu x'^\beta = g_{\alpha\beta}\frac{\partial x^\alpha}{\partial \xi^\mu}\frac{\partial x^\beta}{\partial \xi^\nu}.$$

A trick to remembering this is that the indices must always contract meaning since μ, ν are on the bottom for $g_{\mu\nu}$, the must also be on the bottom, which we have by $\partial_\mu, \partial_\nu$.

Since g' is the freely falling frame, we have

$$g_{\alpha\beta}\frac{\partial x^\alpha}{\partial \xi^\mu}\frac{\partial x^\beta}{\partial \xi^\nu} = \eta_{\mu\nu}.$$

Writing the tensors as matrices, we have inverse matrices

$$\frac{\partial x^\alpha}{\partial \xi^\mu} = \left(\frac{\partial \xi^\alpha}{\partial x^\mu}\right)^{-1}.$$

Multiplying through on our above equation gives

$$\eta\frac{\partial \xi^\mu}{\partial x^\sigma} = g_{\alpha\beta}\frac{\partial x^\alpha}{\partial \xi^\mu}\frac{\partial \xi^\mu}{\partial x^\sigma}\frac{\partial x^\beta}{\partial \xi^\nu} = g_{\alpha\beta}\delta^\alpha_\sigma\frac{\partial x^\beta}{\partial \xi^\nu} = g_{\sigma\beta}\frac{\partial x^\beta}{\partial x^\nu}.$$

Applying the same argument again gives

$$\eta_{\mu\nu}\frac{\partial \xi^\mu}{\partial x^\sigma}\frac{\partial \xi^\nu}{\partial x^\rho} = g_{\sigma\rho}.$$

We are now writing τ instead of λ because the prof's notation got lazy. Applying the chain rule and product rule,

$$0 = \frac{\mathrm{d}^2\xi^\mu}{\mathrm{d}\tau^2} = \frac{\mathrm{d}}{\mathrm{d}\tau}\frac{\mathrm{d}}{\mathrm{d}\tau}\xi^\mu = \frac{\mathrm{d}}{\mathrm{d}\tau}\left(\frac{\mathrm{d}x^\alpha}{\mathrm{d}\tau}\frac{\partial \xi^\mu}{\partial x^\alpha}\right) = \frac{\partial^2\xi^\mu}{\partial x^\alpha \partial x^\beta}\frac{\mathrm{d}x^\alpha}{\mathrm{d}\tau}\frac{\mathrm{d}x^\beta}{\mathrm{d}\tau} + \frac{\mathrm{d}^2 x^\alpha}{\mathrm{d}\tau^2}\frac{\partial \xi^\mu}{\partial x^\alpha}.$$

Of course throughout this we are assuming derivatives freely commute, which indeed follows for smooth maps. We use the same process to strip of the factor of $\partial_\alpha \xi^\mu$ by multiplying through by $\partial_{\xi^\mu} x^\sigma$:

$$0 = \frac{\mathrm{d}x^\sigma}{\mathrm{d}\tau^2} + \frac{\mathrm{d}x^\alpha}{\mathrm{d}\tau}\frac{\mathrm{d}x^\beta}{\mathrm{d}\tau}\frac{\partial x^\sigma}{\partial \xi^\mu}\frac{\partial^2\xi^\mu}{\partial x^\alpha \partial x^\beta}.$$

Wr write

$$\Gamma^\sigma_{\alpha\beta} \coloneqq \frac{\partial^2\xi^\mu}{\partial x^\alpha \partial x^\beta}\frac{\partial x^\sigma}{\partial \xi^\mu}.$$

The answer is then rewritten as

$$0 = \frac{\mathrm{d}^2 x^\sigma}{\mathrm{d}\tau^2} + \Gamma^\sigma_{\alpha\beta}\frac{\mathrm{d}x^\alpha}{\mathrm{d}\tau}\frac{\mathrm{d}x^\beta}{\mathrm{d}\tau}.$$

This is called the *geodesic equation*. We want to rewrite it in terms of a metric to be able to use it more easily. To do this, we have to rewrite $\Gamma^\sigma_{\alpha\beta}$ in terms of the metric

$$\eta_{\mu\nu}\frac{\partial\xi^\mu}{\partial x^\sigma}\frac{\partial\xi^\nu}{\partial x^\rho} = g_{\sigma\rho}.$$

Differentiate with respect to x^λ:

$$\frac{\partial g_{\alpha\beta}}{\partial x^\lambda} = \eta_{\mu\nu}\frac{\partial^2\xi^\mu}{\partial x^\lambda\partial x^\alpha}\frac{\partial\xi^\nu}{\partial x^\beta} + \eta_{\mu\nu}\frac{\partial\xi^\mu}{\partial x^\alpha}\frac{\partial^2\xi^\nu}{\partial x^\lambda\partial x^\beta}.$$

Notice that this is simply

$$\frac{\partial g_{\mu\nu}}{\partial x^\lambda} = \eta_{\mu\nu}\left(\Gamma^\rho_{\lambda\alpha}\frac{\partial\xi^\mu}{\partial x^\rho}\frac{\partial\xi^\nu}{\partial x^\beta} + \Gamma^\rho_{\lambda\beta}\frac{\partial\xi^\mu}{\partial x^\alpha}\frac{\partial\xi^\nu}{\partial x^\rho}\right) = \Gamma^\rho_{\lambda\alpha}g_{\rho\beta} + \Gamma^\rho_{\lambda\beta}g_{\alpha\rho},$$

where the final step follows also by our above formula for $g_{\mu\nu}$.

Note that by definition, the Christofell symbol $\Gamma^\mu_{\alpha\beta}$ is symmetric under the lower indices, so we add the equation to itself under a permutation of indices, which I don't want tow rite. We end up with

$$\partial_\lambda g_{\mu\nu} + \partial_\mu g_{\lambda\nu} - \partial_\nu g_{\mu\lambda} = 2\Gamma^\rho_{\lambda\mu}g_{\rho\nu}.$$

Stripping the metric by multiplying by its inverse:

$$\Gamma^\rho_{\alpha\beta} = \frac{1}{2}g^{\sigma\rho}\left(\frac{\partial g_{\alpha\sigma}}{\partial x^\beta} + \frac{\partial g_{\beta\sigma}}{\partial x^\alpha} - \frac{\partial g_{\alpha\beta}}{\partial x^\sigma}\right).$$

This allows us to rewrite the geodesic equation in terms of the metric only. Since the Minkowski metric has fully dropped out, this olds in *any metric*.

Lecture 6: Covariant derivatives

Thu 18 Sep 2025 12:33

Furthering the topic of covariance, we introduce covariant derivatives. Tensors are automatically invariant under coordinate transformations, but derivatives are not usually tensors. We define a covariant derivative ∇_μ to be a derivative that transforms in the right way. Since ∂_μ transforms properly on scalar functions, we should have $\nabla_\mu f = \partial_\mu f$. We should also have that if S, T are tensors, then $\nabla_\mu(S+T) = \nabla_\mu S + \nabla_\mu T$ (linearity). Finally, it should satisfy the product rule with respect to the tensor product.

Let's see how this acts on a dual vector ω_ν. We should have a tensor $\nabla_\mu\omega_\nu = X_{\mu\nu}$. The only linear maps on dual vectors are derivatives ∂_μ and matrics $C^\alpha_{\mu\nu}$. The most general thing we can have is thus

$$\nabla_\mu\omega_\nu = a\partial_\mu\omega_\nu - C^\alpha_{\mu\nu}\omega_\alpha,$$

for some scalar a. The reason we use a minus sign is so the right side can later become $\Gamma^\alpha_{\mu\nu}$. Condition 1 forces $a = 1$ for scalars, and the product rule forces $a = 1$ for vectors.

Now we consider it on an arbitrary vector v^α. We must have

$$\nabla_\mu v^\alpha = \partial_\mu v^\alpha - B^\alpha_{\mu\beta}v^\beta.$$

We should find a relation between B and C. Interestingly, we can already do this. Since $\omega_\alpha v^\alpha$ is a scalar,

$$\nabla_\mu(\omega_\alpha v^\alpha) = \partial_\mu(\omega_\alpha v^\alpha) = \omega_\alpha \partial_\mu v^\alpha + (\partial_\mu \omega_\alpha)\, v^\alpha.$$

Alterntatively, via the product rule,

$$\nabla_\mu(\omega_\alpha v^\alpha) = (\nabla_\mu \omega_\alpha) v^\alpha + \omega_\alpha \nabla_\mu v^\alpha = \left(\partial_\mu \omega_\alpha - C^\beta_{\mu\alpha}\omega_\beta\right) v^\alpha + \omega_\alpha \left(\partial_\mu v^\alpha - B^\alpha_{\mu\beta} v^\beta\right).$$

These two expressions must be equal. We cancel the prtials to obtain

$$0 = -C^\beta_{\mu\alpha}\omega_\beta v^\alpha + B^\alpha_{\mu\beta}\omega_\alpha v^\beta.$$

Intuitively, $B^\beta_{\mu\alpha} = -C^\beta_{\mu\alpha}$. To prove this, since α, β are dummy indices, we can rename them as

$$C^\beta_{\mu\alpha}\omega_\beta v^\alpha = B^\beta_{\mu\alpha}\omega_\beta v^\alpha.$$

Since this holds for *every* dual vector and *every* vector, this gives our relation. This can be done pointwise by choosing ω, v to pick out individual entries. We now have

$$\nabla_\mu v^\alpha = \partial_\mu v^\alpha + C^\alpha_{\mu\beta} v^\beta.$$

The product rule can generalize to all tensors by taking the product over vectors and duals. Arbitrary tensors generalize by taking the derivative, then adding a term for each index, viewing upper indices as like vectors and lowers like duals.

We are basically correcting the derivative to make it transform like a tensor by adding all the other terms. Any tensor satisfying these properties is a covariant derivative. We specialize by choosing a nice covariant derivative for GR.

We want $C^\rho_{\mu\nu} = C^\rho_{\nu\mu}$, meaning C is symmetric in μ, ν. This is nice for physics reasons, since we can just symmetrize it and antisymmetrize it. The antisymmeterization transforms like a tensor, and if that existed, then that would be an additional field. We have not found this extra field, so we demand it not have an antisymmetric component.

We also require that $\nabla_\mu g_{\alpha\beta} = 0$. We call this metric compatibility. This will help develop the equations of motion from the equivalence principle. This fixes ∇_μ as follows. We have

$$\nabla_\mu g_{\alpha\beta} = \partial_\mu g_{\alpha\beta} - C^\lambda_{\mu\alpha} g_{\lambda\beta} - C^\lambda_{\mu\beta} g_{\alpha\lambda}.$$

We have

$$\partial_\mu g_{\alpha\beta} = C^\lambda_{\mu\alpha} g_{\lambda\beta} + C^\lambda_{\mu\beta} g_{\alpha\lambda}.$$

Since the partial of the metric is $\partial_\mu g_{\alpha\beta} = \Gamma^\lambda_{\mu\alpha} g_{\lambda\beta} + \Gamma^\lambda_{\mu\beta} g_{\alpha\lambda}$, we just obtain

$$C^\lambda_{\alpha\beta} = \Gamma^\lambda_{\alpha\beta}.$$

Here is an example. The metric for flat space in polar coordinates is

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \mathrm{d}r^2 + r^2 \mathrm{d}\theta^2.$$

We thus have $g_{\mu\nu} = \mathrm{diag}(-1, 1, r^2)$. Computing $\Gamma^r_{\theta\theta}$ at a point:

$$\Gamma^r_{\theta\theta} = \frac{1}{2} g^{r\rho}\left(2\partial_\theta g_{\rho\theta} - \partial_\rho g_{\theta\theta}\right) = \frac{1}{2} g^{rr}\left(2\partial_\theta g_{r\theta} - \partial_r g_{\theta\theta}\right) = \frac{1}{2}\frac{1}{r^2}\left(-\partial_r r^2\right) = -r.$$

Lecture 7: Geodesics

Tue 23 Sep 2025 12:30

Consider a vector v^μ tangent to a path $x^\mu(\lambda)$. We have

$$\frac{\mathrm{D}}{\mathrm{d}\lambda} \coloneqq \frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\nabla_\mu.$$

Applying this to v gives

$$0 = \frac{\mathrm{D}}{\mathrm{d}\lambda}v^\mu = \frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\left(\partial_\nu v^\mu + \Gamma^\mu_{\nu\alpha}v^\alpha\right).$$

I skipped something. We want to check as claimed that if two vectors v^μ, w^μ on a path $x^\mu(\lambda)$, then when parallel transported along the path, the angle doesn't change. The $\mathrm{D}/\mathrm{d}\lambda = 0$ is parallel transport or something.

$$\frac{\mathrm{D}}{\mathrm{d}\lambda}\left(v^\mu w_\mu\right) = 0.$$

If they are parallel transported, then

$$\frac{\mathrm{D}}{\mathrm{d}\lambda}v^\mu = \frac{\mathrm{D}}{\mathrm{d}\lambda}w^\mu = 0.$$

Computing:

$$\frac{\mathrm{D}}{\mathrm{d}\lambda}(v^\mu w_\mu) = \frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\nabla_\nu(v^\mu w_\mu).$$

By the product rule, we see

$$= \frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\left(\nabla_\nu(v^\mu)w_\mu + v^\mu\nabla_\nu w_\nu\right).$$

The first term goes to 0 by hypothesis, and the second term has its index raised by the metric

$$= \frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}(v^\mu\nabla_\nu g_{\mu\alpha}v^\alpha) = \frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}v^\mu\left(g_{\mu\alpha}\nabla_\nu v^\alpha + g_{\mu\alpha}\nabla_\nu v^\alpha\right).$$

The second term is 0 by hypothesis, and the first is 0 since the covariant derivative of the metric vanishes by construction.

What happens if we require $\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}$ be parallel transported, meaning the angle we are traveling in is invariant along the curve? Then

$$0 = \frac{\mathrm{D}}{\mathrm{d}\lambda}\left(\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\right) = \frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\nabla_\nu\left(\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\right) = \frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\left(\frac{\partial}{\partial x^\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda} + \Gamma^\mu_{\nu\sigma}\frac{\mathrm{d}x^\sigma}{\mathrm{d}\lambda}\right) = \frac{\mathrm{d}}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda} + \Gamma^\mu_{\nu\sigma}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\sigma}{\mathrm{d}\lambda}.$$

The last equality in the first term is chain rule. This is simply the geodesic equation:

$$0 = \frac{\mathrm{d}^2x^\mu}{\mathrm{d}\lambda^2} + \Gamma^\mu_{\nu\sigma}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}.$$

Thus, the geodesic equation is simply the requirement that the derivative be parallel transported.

We would like to simplify computation of Christoffel symbols. Recall that given a distance $\mathrm{d}x^\mu$, the change in proper time is

$$\mathrm{d}\tau^2 = -g_{\mu\nu}\mathrm{d}x^\mu\mathrm{d}x^\nu = -g_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\mathrm{d}\lambda^2.$$

We write

$$\tau = \int \mathrm{d}\tau = \int \sqrt{-g_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}}\,\mathrm{d}\lambda.$$

We are going to extremize τ by taking the variance $\delta\tau$ with respect to x^μ:

$$\delta\tau = \int \frac{1}{2}\left(-g_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\right)^{-1/2}\left(\left(-\frac{\partial}{\partial x^\sigma}g_{\mu\nu}\right)\delta x^\sigma\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda} - g_{\mu\nu}\frac{\mathrm{d}\delta x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda} - g_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}\delta x^\nu}{\mathrm{d}\lambda}\right)\mathrm{d}\lambda.$$

Of course, $\delta x^\mu = \delta(x^\mu)$. Since μ, ν are contracted with the metric, they are dummy variables, so we can rename them

$$\delta\tau = \int \frac{1}{2}\left(-g_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\right)^{-1/2}\left(\left(-\frac{\partial}{\partial x^\sigma}g_{\mu\nu}\right)\delta x^\sigma\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda} - g_{\mu\nu}\frac{\mathrm{d}\delta x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda} - g_{\mu\nu}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\frac{\mathrm{d}\delta x^\mu}{\mathrm{d}\lambda}\right)\mathrm{d}\lambda.$$

Of course, the last two terms are now the same, so since derivatives commute,

$$\delta\tau = \int \frac{1}{2}\left(-g_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\right)^{-1/2}\left(\left(-\frac{\partial}{\partial x^\sigma}g_{\mu\nu}\right)\delta x^\sigma\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda} - 2g_{\mu\nu}\frac{\mathrm{d}\delta x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\right)\mathrm{d}\lambda.$$

Note that

$$\left(-g_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\right)^{-1/2} = \left(\frac{\mathrm{d}\tau^2}{\mathrm{d}\lambda^2}\right)^{-1/2} = \frac{\mathrm{d}\lambda}{\mathrm{d}\tau}.$$

We then have

$$\begin{aligned}\delta\tau &= \int \frac{1}{2}\left(\left(-\frac{\partial}{\partial x^\sigma}g_{\mu\nu}\right)\delta x^\sigma\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda} - 2g_{\mu\nu}\frac{\mathrm{d}\delta x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\right)\frac{\mathrm{d}\lambda}{\mathrm{d}\tau}\mathrm{d}\lambda \\ = \delta\tau &= \int \frac{1}{2}\left(\left(-\frac{\partial}{\partial x^\sigma}g_{\mu\nu}\right)\delta x^\sigma\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda} - 2g_{\mu\nu}\frac{\mathrm{d}\delta x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\right)\left(\frac{\mathrm{d}\lambda}{\mathrm{d}\tau}\right)^2\mathrm{d}\tau.\end{aligned}$$

We have

$$\frac{\partial x^\mu}{\partial\lambda} = \frac{\partial x^\mu}{\partial\tau}\frac{\partial\tau}{\partial\lambda},$$

giving

$$\delta\tau = \int \frac{1}{2}\left(\left(-\frac{\partial}{\partial x^\sigma}g_{\mu\nu}\right)\delta x^\sigma\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau}\left(\frac{\mathrm{d}\tau}{\mathrm{d}\lambda}\right)^2 - 2g_{\mu\nu}\frac{\mathrm{d}\delta x^\mu}{\mathrm{d}\tau}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau}\left(\frac{\mathrm{d}\tau}{\mathrm{d}\lambda}\right)^2\right)\left(\frac{\mathrm{d}\lambda}{\mathrm{d}\tau}\right)^2\mathrm{d}\tau.$$

The derivatives cancel, yielding

$$\delta\tau = \int \frac{1}{2}\left(\left(-\frac{\partial}{\partial x^\sigma}g_{\mu\nu}\right)\delta x^\sigma\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} - 2g_{\mu\nu}\frac{\mathrm{d}\delta x^\mu}{\mathrm{d}\tau}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau}\right)\mathrm{d}\tau.$$

Apparently we can just write this as

$$\delta\left(\frac{1}{2}\int -g_{\alpha\beta}\frac{\mathrm{d}x^\alpha}{\mathrm{d}\tau}\frac{\mathrm{d}x^\beta}{\mathrm{d}\tau}\,\mathrm{d}\tau\right).$$

This is a nicer variational action to use, since the square root is gone. Interestingly, the integrand is 1 by a previous equation. This is not quite the version we want to use, since we have a derivative of the variation. We want it to just be an integral times $\mathrm{d}x^\mu$. The EOM is then integrand equal to 0.

To fix this, we do some integration by parts, and the boundary vanishes since there is no variation at the boundary.

Remark 3.0.2. the δ is just a derivative. but with... different symbols.

I don't know why we're going on the tangent we can just recall the geodesic equation and definition of the Christoffel symbol in terms of the metric:

$$\delta\tau = \int g_{\sigma\rho}\left(\frac{\mathrm{d}^2 x^\rho}{\mathrm{d}\tau^2} + \Gamma^\rho_{\alpha\beta}\frac{\mathrm{d}x^\alpha}{\mathrm{d}\tau}\frac{\mathrm{d}x^\beta}{\mathrm{d}\tau}\right)\delta x^\sigma\,\mathrm{d}\tau.$$

The extremitization is given by just taking the integrand and dropping the δx^μ by the fundamental theorem of the calculus of variations. Wow, the equation of motion is just the geodesic equation.

If $g_{\mu\nu}$ is arbitrary, then Christoffel symbol computations are just brute force. However, if the metric has symmetries (eg. is symmetric), then plugging into the action principal can simplify computations.

So

$$0 = 2g_{\sigma\beta}\left(\frac{\mathrm{d}^2\beta}{\mathrm{d}\tau^2} + \Gamma^\beta_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau}\right).$$

Let's do an example with flat space in polar coordinates $\mathrm{d}s^2 = -\mathrm{d}t^2 + \mathrm{d}r^2 + r^2\mathrm{d}\theta^2$.

$$L \equiv \frac{1}{2}\int -g_{\alpha\beta}\frac{\mathrm{d}x^\alpha}{\mathrm{d}\tau}\frac{\mathrm{d}x^\beta}{\mathrm{d}\tau}\,\mathrm{d}\tau = \frac{1}{2}\int\left(\frac{\mathrm{d}t}{\mathrm{d}\tau}^2 - \frac{\mathrm{d}r}{\mathrm{d}\tau}^2 - \frac{\mathrm{d}r}{\mathrm{d}\tau}^2\frac{\mathrm{d}\theta}{\mathrm{d}\tau}^2\right)\mathrm{d}\tau.$$

Lecture 8: Expanding Universe

Tue 30 Sep 2025 12:31

The simplest metric we can use with an expanding component is

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + a^2(t)\left(\mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2\right).$$

This is the simplest case of an FRW universe. Last time we worked through the geodesic equation using the action principle, giving us

$$0 = \frac{\mathrm{d}}{\mathrm{d}\lambda}\frac{\partial L}{\partial q'} - \frac{\partial L}{\partial q},$$

$$q = t, \qquad q = x^i, \qquad q' = \frac{\mathrm{d}q}{\mathrm{d}\lambda}, \qquad \dot{f} = \frac{\mathrm{d}f}{\mathrm{d}t}.$$

$$L = \frac{1}{2}\int -g_{\mu\nu}x^{\mu'}x^{\nu'}\,\mathrm{d}\lambda.$$

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}a(t) = \dot{a}(t)\frac{\mathrm{d}t}{\mathrm{d}\lambda}.$$

We then found

$$0 = t'' + a\dot{a}\left(\frac{t'}{a}\right)^2.$$

$$0 = \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(a^2(t)x_i'\right) = a^2(t)\left(x_i'' + 2\frac{\dot{a}(t)}{a(t)}t'x_i'\right).$$

Staring at this somehow gives us the Christoffel symbol in t

$$\Gamma^t_{\alpha\beta} = a\dot{a}\delta_{\alpha\beta}.$$

This is because the t'' equation is the Geodesic equation

$$t'' + a\dot{a}\left(\frac{t'}{a}\right)^2 = q'' + \Gamma^q_{\alpha\beta}x^{\alpha'}x^{\beta'}.$$

This is equal probably because they're both 0. We then read it off since they're the same and t'' becomes q''. For $\Gamma^i_{\alpha\beta}$ in x^i,

$$\Gamma^i_{\alpha\beta} = \frac{\dot{a}}{a}\delta^j_i.$$

Ok now we want to look at a photon travelling through the expanding universe. Since we cannot use proper time τ, we use an affine parameter $\lambda' = A\lambda + B$. We have a lot of freedom for choosing this affine parameter, in particular the scaling factor A. We are going to choose λ such that in a fixed reference frame of our choice, $p^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}$, for p^μ photon momentum. Such a frame exists. We will always assume the affine parameter for massless particles is chosen in this way.

The energy is just $E = p^0 = \frac{\mathrm{d}x^0}{\mathrm{d}\lambda}$. We need to solve the Geodesic equation for this, and then we need to interpret the meaning of this physically. We will do the latter by choosing locally inertial coordinates somewhere else, and then use general covariance.

The energy of a photon has observer dependence (doppler shift). Given a 4-momentum U^μ of our observer (reference frame), what is the energy of the photon p^μ? We write p^0 in the observer rest frame in a coordinate invariant way. We do this by general covariance, so first we write it in locally inertial coordinates. We have the 4-momentum of the observer and photon. We can take the inner product

$$-U_\mu p^\mu = -U^\mu p^\nu g_{\mu\nu}.$$

In the observer's rest frame, $U^\mu = (1,0,0,0)$, so we only get a contribution when $\mu = 0$:

$$-p^\nu g_{0\nu} = p^0.$$

This is the energy of the photon, and is manifestly coordinate invariant since $E = -U_\mu p^\mu$.

Now we solve the geodesic equation. Since massless particles travel along right rays, $\mathrm{d}s^2 = 0$. We then have

$$0 = -\left(\frac{\mathrm{d}t}{\mathrm{d}\lambda}\right)^2 + a^2(t))\left(\frac{\mathrm{d}x}{\mathrm{d}\lambda}\right)^2.$$

Whenever we want to solve for the geodesic equation of a massless particle, start here, since like its kinda like the Geodesic equation. We assume the photon is traveling in the x direction only in the λ reference frame by rotational invariance. This is why there is only a contribution from x. Solving this equation gives

$$\frac{\mathrm{d}t}{\mathrm{d}\lambda} = a(t)\frac{\mathrm{d}x}{\mathrm{d}\lambda} \iff \frac{\mathrm{d}x}{\mathrm{d}\lambda} = \frac{1}{a(t)}\frac{\mathrm{d}t}{\mathrm{d}\lambda}.$$

We see that geodesic equation is of the form

$$0 = t'' + a\dot{a}(x')^2 = t'' + \frac{\dot{a}}{a}(t')^2.$$

This is solved by $t' = \frac{\omega_0}{a}$ for ω_0 some integration constant.

Recall that the energy of the photon was

$$E = -U_\mu p^\mu = -U_\mu (x')^\mu.$$

We choose coordinates $U^\mu = (1, 0, 0, 0)$ because the Earth is not moving at the speed of light, and because we can take the entire universe to be moving with respect to us (we are moving slowly through it). So this becomes

$$E = g_{00} U^0 t' = \frac{\omega_0}{a}.$$

In particular, the energy of a photon *decreases* as the universe expands. This makes sense since the energy is inversely proporitional to the photon, and as it expands, the wavelength expands. The redshift is just

$$Z = \frac{\Delta\lambda}{\lambda_i} = \frac{\lambda_f - \lambda_i}{\lambda_i} = \frac{a(t_f)}{a(t_i)} - 1.$$

We choose $a(t)$ for $t =$ today to be 1, so we simply have

$$Z = \frac{1}{a(t_i)} - 1.$$

In reality, the universe has things in it. They all have the same form for their stress tensor, which is the perfect fluid stress tensor

$$T_{\mu\nu} = (p + \rho)\, U_\mu U_\nu + p g_{\mu\nu}.$$

Here, ρ is the energy density and p is the pressure. We can find p given ρ for matter, photons, and dark energy, which are the three things in the universe.

For a single particle $x^\mu(\lambda)$, apparently

$$T^{\mu\nu} = \int p^\mu \frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda} \delta^3(x - x(\lambda))\, \mathrm{d}\lambda.$$

If we choose the rest frame of the particle, then

$$T^{00} = p^0.$$

The formula is manifestly covariant, so we now compute the pressure. Apparently $T^i_j = p\delta^i_j$. Apparently the answer depends on the temperature of the matter, but in space that is basically 0. For cold matter, $p = 0$. So the equation of state

$$w = \frac{p}{\rho} = 0$$

for cold matter. For photons, $w = 1/3$, so $\rho = 3p$. This follows because p^2 (momentum) is 0 for massless particles, since the square of the 4-momentum is the rest mass.

For dark energy, we assume dark energy is a cosmological constant, meaning $T^{\mu\nu}$ is proportional to $-\eta^{\mu\nu}$. This is the only thing we can have without breaking Lorentz invariance. We then see that $w = -1$.

So now, how does ρ change with the expansion of the universe? Intuitively, $\rho \sim 1/a^3$ for matter, since volume is proportional to length cubed. For photons, the wavelength also gets redshifted, so we expect it to be $\rho \sim 1/a^4$. The cosmological constant is... constant... $T^{\mu\nu} = \Lambda\eta^{\mu\nu}$ and Λ is independent.

Without general relativity, $\partial_\mu T^{\mu\nu} = 0$. This isn't a tensor, so we have to take the covariant derivative to make it a tensor:

$$\nabla_\mu T^{\mu\nu} = 0.$$

This can be solved two different ways. It is easier to solve it if we lower ν:

$$\nabla_\mu T^\mu_\nu.$$

Of course, we can always relate these by $g_{\mu\nu}$. We have

$$\nabla_\mu T^\mu_\nu = \partial_\mu T^\mu_\nu + \Gamma^\nu_{\mu\alpha} T^\alpha_\nu - \Gamma^\alpha_{\mu\nu} T^\mu_\alpha.$$

We found peviously that in the FRW metric,

$$\Gamma^t_{ij} = a\dot{a}\delta_{ij}, \qquad \Gamma^i_{tj} = \frac{\dot{a}}{a}\delta^i_j,$$

and all others are 0. Plugging this in,

$$0 = \partial_0 T^0 + \partial_i T^i_\nu + \Gamma^0_{0\alpha} T^\alpha_0 + \Gamma^i_{i\alpha} T^\alpha_0 - \Gamma^0_{\mu 0} T^\mu_0 - ?.$$

Apparently most vanish because of some magic by seeing that off-diagonals vanish on the Christoffel symbols. The nonvanishing terms are

$$= \partial_t T^i_t + \Gamma^i_{it} T^t_t - \Gamma^i_{jt} T^j_i = .$$

We use $T^\mu_\nu = \mathrm{diag}(\rho, p, p, p)$ and our definition of the Christoffel symbols to see that this is

$$-\dot{\rho} - \frac{\dot{a}}{a}\delta^i_i \rho - \frac{\dot{a}}{a}\delta^i_j w\rho\delta^j_i = -\left(\dot{\rho} + 3\frac{\dot{a}}{a}(1+w)\rho\right).$$

This is solved by

$$\rho(t) = \frac{\rho_0}{(a(t))^{3(1+w)}}.$$

Lecture 9: Curvature

Thu 02 Oct 2025 12:32

We start with the Riemann tensor. We are given a metric $g_{\mu\nu}$ on a manifold M. We want to find combinations of its derivatives that are tensors. We could do $\nabla_\mu g_{\rho\sigma}$, which is the simplest tensor. Unfortunately, this is 0 by definition, so this is useless.

We use a trick to do this. Take a vector V^μ which is a rank 1 tensor. Then $\nabla_\mu \nabla_\nu V^\alpha$ is a rank 3 tensor. We are supposed to only use the metric though, but it turns out if we instead take the commutator

$$\left[\nabla_\mu, \nabla_\nu\right] V^\alpha = \nabla_\mu \nabla_\nu V^\alpha - \nabla_\nu \nabla_\mu V^\alpha$$

is a tensor which is *invariant* under the choice of V. This will evaluate to $R^\alpha_{\rho\mu\nu} V^\rho$. The left side is manifestly a tensor, since the derivative terms cancel (cuz they commute), and the part with only one derivative vanishes since we can pass to a frame where ∂g vanishes, so we just get the part with the Christoffels. Now we must just show this equality.

We work this out.

$$\nabla_\mu \left(\nabla_\nu V^\rho\right) = \partial_\mu \left(\nabla_\nu V^\rho\right) + \Gamma^\rho_{\mu\alpha}(\nabla_\nu V^\alpha) - \Gamma^\alpha_{\mu\nu}(\nabla_\alpha V^\rho) - (\mu \leftrightarrow \nu)$$

$$= \partial_\mu \partial_\nu V^\rho + \partial_\mu \Gamma^\rho_{\nu\beta} V^\beta + \Gamma 3^\rho_{\nu\beta} \partial_\mu V^\beta + \Gamma^\rho_{\mu\alpha}\left(\partial_\nu V^\alpha + \Gamma^\alpha_{\nu\beta} V^\beta\right) - \Gamma^\alpha_{\mu\nu}\left(\partial_\alpha V^\rho + \Gamma^\rho_{\alpha\beta} V^\beta\right) - (\mu \leftrightarrow \nu).$$

Some stuff cancels giving

$$= \left(\partial_\mu \Gamma^\rho_{\nu\beta} + \Gamma^\rho_{\mu\alpha}\Gamma^\alpha_{\nu\beta} - (\mu \leftrightarrow \nu)\right) V^\beta.$$

We see that the term in the parenthases is the rank 4 tensor $R^\rho_{\beta\mu\nu}$. Explicitly,

$$R^\rho_{\beta\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\beta} - \partial_\nu \Gamma^\rho_{\mu\beta} + \Gamma^\rho_{\mu\alpha}\Gamma^\alpha_{\nu\beta} - \Gamma^\rho_{\nu\alpha}\Gamma^\alpha_{\mu\beta}.$$

There is another way to define it. Suppose we have a vector V^α which we parallel transport around flat space. On a *curved* manifold, if we parallel transport, then when we come back to the same spot it could be pointing a *different* direction. In particular, if there is *any* curvature, the vector will always rotate (as long as we walk in more than 1 direction). Let's say we walk an infintesimal amount towards A^α then B^α, then $-A^\alpha$ then $-B^\alpha$. In curved space, V^α undergoes some linear map $M^\alpha_\nu(A, B)V^\nu$. We Taylor expanded in small A, B. I don't care. The first nonzero term we get is

$$A^\mu B^\nu R^\alpha_{\beta\mu\nu}.$$

Wow the Riemann tensor does curvy stuff. However, it was only the leading term of the Taylor expansion. We will spend the remainder of this lecture showing it captures *all* the invariant curvature of the manifold.

Clearly, $R^\rho_{\sigma\mu\nu} = 0$ when $g_{\mu\nu} = \eta_{\mu\nu}$ *everywhere* in some coordinate system (flat spacetime). One direction is easy, since derivatives of $\eta_{\mu\nu}$ are 0.

Start by choosing $p \in M$ and a dual vector ω_μ at p ($\omega \in T_p M$). We want to promote ω_μ to a function $\omega_\mu(x)$, meaning it is a dual vector at any point p. We choose a path $x^\mu(\lambda)$ and parallel transport it along the path. This gives $\omega_\mu(x^\mu(\lambda))$ by parallel transport (recall that is given by solving $0 = \frac{\mathrm{D}}{\mathrm{d}\lambda}\omega_\mu = \frac{\mathrm{d}x^\nu}{\lambda}\nabla_\nu \omega_\mu$). This only promotes ω along the given curve, so we choose all possible curves. It's not clear this is well-defined, since two paths could be acted on by ω differently at the same point.

However, recall that the differential in a transport around a square $\delta V^\alpha = R^\alpha_{\rho\mu\nu}A^\mu B^\nu V^\rho$. By assumption, this vanishes. Therefore, the differential in the change of V^α along different parallel transports vanishes, so parallel transporting along different curves is the same (parallel transport forward along the first and backwards along the second, it has to vanish, so they are the same). We thus have a well-defined promotion of ω to a dual vector at any point $\omega_\mu(x)$. Now note that if

$$\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\nabla_\nu \omega_\mu(x) = 0 \implies \nabla_\nu \omega_\mu = 0,$$

since this must hold for *any* possible path, so dotting the derivative into the tensor always vanishes. We thus have

$$0 = \partial_\mu \omega_\nu - \Gamma^\alpha_{\mu\nu}\omega_\alpha.$$

Antisymmetrize on $\mu\nu$ (swap μ and ν then subtract from itself):

$$0 = -\partial_\nu \omega_\mu + \Gamma^\alpha_{\nu\mu}\omega_\alpha + \partial_\mu \omega_\nu - \Gamma^\alpha_{\mu\nu}\omega_\alpha = \partial_\mu \omega_\nu - \partial_\nu \omega_\mu.$$

This is because the Christoffel is symmetric on the two indices. Apparently this is similar to the electromagnetic field in terms of the Gauge potential A_μ:

$$\partial_\mu A_\nu - \partial_\nu A_\mu = F_{\mu\nu}$$

for $F_{\mu\nu}$ the field strength. If the field strength is 0, then there is no field, so A_μ is a *pure gauge*, so it is the gradient of a potential function $A_\mu = \partial_\mu \Phi$. This is of course because $\partial_\mu \omega_\nu - \partial_\nu \omega_\mu$ is something we remember from calc 3 as a test for when vector fields are conservative. Therefore, $\omega_\mu = \partial_\mu \Phi$ for some Φ.

Choose locally inertial coordinates at $p \in M$, such that $g_{\mu\nu} \to \eta_{\mu\nu}$ at p. We want to keep track of the coordinate transformation, so write $g_{\mu\nu} = \eta_{\alpha\beta}\omega^{(\alpha)}_\mu \omega^{(\beta)}_\nu$. This gives four 4-vectors $\omega^{(\alpha=1)}_\mu$, and since each α gives a four vector, we have $\omega_\mu^{(\alpha)} = \partial_\mu \Phi^{(\alpha)}$. Since this extends *everywhere* along the manifold, we are done since we can always write

$$g_{\mu\nu} = \eta_{\alpha\beta} \frac{\partial \Phi^{(\alpha)}}{\partial x^\mu} \frac{\partial \Phi^{(\beta)}}{\partial x^\nu}.$$

We have the new coordinate system $(x')^\alpha = \Phi^{(\alpha)}(x)$, which turns the coordinate system into Minkowski everywhere.

Lecture 10: Integration

Tue 07 Oct 2025 12:31

He gave some advice on what we really need to know about tensors. The main thing to remember is the transformation rule

$$M_\mu \to M'_\mu = M_\alpha \frac{\mathrm{d}x^\alpha}{\mathrm{d}(x')^\mu}$$

$$M^\mu \to (M')^\mu = M^\alpha \frac{\mathrm{d}(x')^\mu}{\mathrm{d}x^\alpha}$$

and the raising and lowering

$$M_\mu = M^\alpha g_{\alpha\mu}, \qquad M^\mu = M_\alpha g^{\alpha\mu}.$$

Of course we may iterate the transformation rule when there are more indices; for example

$$M_{\mu\nu} \to M'_{\mu\nu} = M_{\alpha\beta} \frac{\mathrm{d}x^\alpha}{\mathrm{d}(x')^\mu} \frac{\mathrm{d}x^\beta}{\mathrm{d}(x')^\nu}.$$

We also recall that all derivatives commute and that derivatives are vectors and $M_{\alpha\beta}$ is a scalar value acting on it here. The only other main thing to remember is the definition of ∇_μ as ∂_μ plus some Christoffel, the geodesic equation, and the fact that we can always take locally inertial coordinates $g_{\mu\nu}|_p = \eta_{\mu\nu}$ where the first partials vanish.

The Ricci tensor is given by

$$R_{\mu\nu} = R^\lambda_{\mu\lambda\nu},$$

and the Ricci scalar is

$$R = R^\mu_\mu = g^{\mu\nu} R_{\mu\nu}.$$

Define

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R.$$

This is called the *Einstein tensor*, and it satisfies

$$\nabla^\mu G_{\mu\nu} = 0.$$

We now want to do integration. At each point there exists locally inertial coordinates, so we choose

$$g_{\mu\nu} \to \eta_{\mu\nu} = g_{\alpha\beta} \frac{\partial x^\alpha}{\partial (x')^\mu} \frac{\partial x^\beta}{\partial (x')^\nu}.$$

Thus for any metric, we can always work out the jacobian to pass to flat coordinates, giving

$$\int f(x') \, \mathrm{d}^n x' = \int \left| \frac{\partial (x')^\mu}{\partial x^\nu} \right| f(x) \, \mathrm{d}^n x.$$

We don't want to have to compute the determinant every time, so we write it in terms of the metric. Take the determinant of both sides of the metric transformation $g \to \eta$:

$$-1 = \det \eta_{\mu\nu} = \det \left(g_{\alpha\beta} \frac{\partial x^\alpha}{\partial (x')^\mu} \frac{\partial x^\nu}{\partial (x')^\nu} \right) = \det g_{\mu\nu} \det \left(\frac{\partial x^\alpha}{\partial (x')^\beta} \right)^2.$$

We thus have that the determinant we are interested in is simply

$$\left| \frac{\partial (x')^\beta}{\partial x^\alpha} \right| = \sqrt{-\det g_{\mu\nu}}.$$

We choose to write $g = -\det g_{\mu\nu}$. Thus $\mathrm{d}^n x' = \mathrm{d}^n x \sqrt{|g|}$.

Let's do an example. On the 2-sphere, $g_{\mu\nu} = \mathrm{diag}(1, \sin^2\theta)$. We have $g = \sin^2\theta$, so integrals on the sphere are given by integrating over $\mathrm{d}\theta \mathrm{d}\phi \sin\theta f(\theta, \phi)$.

It will be useful to define integration using differential forms, rather than relying on the presence of a metric. Define the Levi-Civita *tensor* $\varepsilon_{\mu_1 \dots \mu_k}$ as the Levi-Civita *symbol* times the square root of g:

$$\varepsilon_{\mu_1 \dots \mu_k} = \widetilde{\varepsilon}_{\mu_1 \dots \mu_k} \sqrt{g}.$$

This is a form. Let's recall that a tensor T is

$$T = T_{\mu_1 \dots \mu_n} \mathrm{d}x_1^{\mu_1} \otimes \dots \otimes \mathrm{d}x_n^{\mu_n}.$$

Idk what happened but we introduce the wedge product $\wedge$. We defined it as like $x \wedge y = x \otimes y - y \otimes x$, but its just $\otimes$ quotiented by alternating. Then ε in spherical coordinates somehow looks like

$$\int \varepsilon = \int \sin\theta \, (\mathrm{d}\theta \otimes \mathrm{d}\phi - \mathrm{d}\phi \otimes \mathrm{d}\theta) = 0.$$

So we just replace it with $\mathrm{d}\theta \wedge \mathrm{d}\phi$ and integrate

$$\int \varepsilon = \int \sin\theta \, \mathrm{d}\theta \wedge \mathrm{d}\phi := \int \sin\theta \, \mathrm{d}\theta \mathrm{d}\phi.$$

This is how we define integtation over forms or something. A p-form is then just of the form

$$A = A_{\mu_1 \dots \mu_k} \mathrm{d}x_1^{\mu_1} \wedge \dots \wedge \mathrm{d}x_k^{\mu_k}.$$

We then have

$$\int A = \int A_{\mu_1 \dots \mu_k} \, \mathrm{d}x_1 \dots \mathrm{d}x_k.$$

We can integrate p-forms without appealing to the metric or computing jacobians. Woo!

We can also compute derivatives of forms. We want $\partial_\alpha A_{\mu_1 \dots \mu_k}$ to be a $p+1$ form, so we just antisymmetrize on everything and we call it the exterior derivative. We call this $\mathrm{d}A$. If we do this twice, then since partials commute we actually have $\mathrm{d}^2 A = 0$.

We say A is *exact* if $A = \mathrm{d}B$ for some form B, and A is *closed* if $\mathrm{d}A = 0$. Exact implies closed, but not the converse since we could have A be symmetric in its indices. It would be nice to know which ones are exact though. Next time, we will see that whether or not closed forms are exact depends on the *topology* of the manifold. Oh also Stokes' theorem: on a manifold M and with a form ω,

$$\int_M \mathrm{d}\omega = \int_{\partial M} \omega.$$

Lecture 11: Maximally symmetric spaces

Tue 21 Oct 2025 12:32

Recall that there were only 6 maximally symmetric spaces. They must have a constant Ricci scalar R and either Euclidean or Lorentzian metric signature. This means we have six different possible spaces:

1. S^d

2. $\mathbb{R}^d$

3. $\mathbb{R}^d$ with negative curvature,

all of which can be summarized as

$$\mathrm{d}s^2 = \frac{\mathrm{d}r^2}{1-\kappa r^2} + r^2 \mathrm{d}\Omega_{d-1}^2,$$
$$\mathrm{d}\Omega_d^2 = \mathrm{d}x_1^2 + \cdots + \mathrm{d}x_d^2.$$

Any choice of κ gives a maximally symmetric space. Here $\kappa = 1$ is S^d, $\kappa = 0$ is $\mathbb{R}^d$, and $\kappa = -1$ is hyperbolic space $\mathbb{H}^d$. These are the euclidean spaces. Adding a time dimension and space expansion gives

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + R^2(t)\left(\frac{\mathrm{d}r^2}{1-\kappa r^2} + r^2\mathrm{d}\Omega_d^2\right).$$

The $R(t)$ is proporitional to the scale factor $a(t)$ which we used previously in the expanding universe metric. The main difference is that $R(t)$ has units of length, whereas $a(t)$ is dimensionless. We fix some time t_0 and set $a(t) = R(t)/R(t_0)$.

In our universe, κ is nearly 0. We are pretty confident that $\kappa = 0$, which is why our inverse seems like flat space, so we just need to measure $a(t)$. Why is $\kappa = 0$ in our universe?

Lets go back and look at the Lorentzian maximally symmetric spaces. One is just Minkowski space where $\kappa = 0$. More generally,

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \ell^2\cosh^2(t/\ell)\left(\frac{\mathrm{d}r^2}{1-\frac{r^2}{\ell^2}} + r^2\mathrm{d}\Omega_{d-2}^2\right).$$

This is called de Sitter space. This is a special case of the expanding universe metric where $\ell^2\cosh^2(t/\ell) = a^2(t)$ and $\kappa = \frac{1}{\ell^2}$. This is important because the expanding universe is not maximally symmetric on all time but only on time slices, whereas de Sitter space is globally maximally symmetric.

De Sitter space is a great approximation to our universe. In this case, ℓ is the analogue of the radius of the universe, and for our universe to be de Sitter we would need to take ℓ to be massive. Our universe is closer to de Sitter space than Minkowski space.

Hyperbolic cosine blows up exponentially fast

$$a(t) = \ell\cosh^2(t/\ell) = \frac{\ell}{2}\left(e^{t/\ell} + e^{-t/\ell}\right).$$

If $t \gg \ell$ then approximately $a(t) \approx \frac{\ell}{2}e^{-t/\ell}$, and we can write

$$\mathrm{d}s^2 \approx -\mathrm{d}t^2 + e^{2(t-t_0)/\ell}\left(\frac{\mathrm{d}r^2}{1-\frac{1}{\ell}e^{-t_0^2/\ell^2}r^2} + r^2\mathrm{d}\Omega_{d-2}^2\right).$$

So if we observe today in natural units then it looks like $\frac{1}{\ell^2}e^{-t_0^2/\ell^2}$ which looks Minkowski.

Lecture 12: Conformal diagrams

Thu 23 Oct 2025 12:31

We use the change of variables $u = t - r$, $v = t + r$, $U = \arctan u$, and $V = \arctan v$. Then

$$\mathrm{d}s^2 = -\mathrm{d}u\mathrm{d}v + \left(\frac{v-u}{2}\right)^2 \mathrm{d}\Omega = -\left(\frac{\mathrm{d}U}{\cos^2 U}\right)\left(\frac{\mathrm{d}V}{\cos^2 V}\right) + \frac{(\tan V - \tan U)^2}{4}\mathrm{d}\Omega^2.$$

Trig identities give

$$= \frac{1}{\cos^2 U \cos^2 V}\left(-\mathrm{d}U\mathrm{d}V + \sin^2(V-U)\mathrm{d}\Omega^2\right).$$

We can draw cool diagrams.

Chapter 4

Einstein's equation

We will now shift our focus. We will discuss Einstein's equation, which tells us what the metric is *as a function of time.* This is much richer since the metric has to be governed by the existence of matter and energy, and the metric should respond to the existence of this.

Lecture 13: Einstein's equation

Tue 28 Oct 2025 12:34

We want to promote Newton's equation $\vec{\partial} \cdot \vec{\partial} \cdot \Phi = 4\pi G_N \rho$ to Einstein's equation somehow. Define

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}.$$

Then Biachi's identity gives

$$\nabla_\mu G^{\mu\nu} = 0.$$

I zoned out and then we got

$$G_{\mu\nu} = 8\pi G_N T_{\mu\nu} = \kappa T_{\mu\nu}, \qquad \kappa = 8\pi G_N.$$

The action principle for general relativity is

$$S_{\mathrm{EH}} = \int R\sqrt{-g}\,\mathrm{d}^4 x.$$

This is called the Einstein-Hilbert action. The action principle somehow gives the Lagrangian

$$\sqrt{-g}\,(G_{\mu\nu} - 8\pi G_N T_{\mu\nu})\,\delta g^{\mu\nu}.$$

Therefore, we obtain Einstein's equation.

Lecture 14: Black holes

Thu 30 Oct 2025 12:33

Ok so the action is like

$$S = \frac{1}{16\pi G_N} \int (R - 2\Lambda)\sqrt{-g}\,\mathrm{d}^4 x + S_{\mathrm{SM}}.$$

The Λ is a cosmological constant. Now the S_{SM} contains the higher order terms of the Ricci scalar $(R^2, R^3, \ldots)$, but we think that their coefficients are small enough that the EFT given by S matches observations to high precision. Here G_N is the gravitationa coupling constant. Using GR as an EFT is the only way we can make it work as a quantum theory.

What if instead of $R - 2\Lambda$, we had some arbitrary $f(R)$? We can expand this as a Taylor series to obtain an expansion in powers of R. The extra terms would be dominated by small numbers so we recover Einstein's equation. So if we have some $f(R)$, then their coupling to the standard model is tiny.

Chapter 5

Black holes

Ok so recall Einstein's equation

$$G_{\mu\nu} = 8\pi G_N T_{\mu\nu}.$$

We are going to solve the simplest black hole: the Schwarzschild black hole. Consider the simple case where $8\pi G_N T_{\mu\nu} = 0$. The equation $G_{\mu\nu} = 0$ is a second order nonlinear differential equation, and is nontrivial to solve. In order to actually solve these equations, we will consider highly symmetric scenarios.

Start by assuming rotational invariance and time invariance. Minkowski space is rotationally symmetric, where we write the metric as

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \mathrm{d}r^2 + r^2\mathrm{d}\Omega_2^2 = -\mathrm{d}t^2 + \mathrm{d}r^2 + r^2\left(\mathrm{d}\theta^2 + \sin^2\theta\mathrm{d}\phi^2\right).$$

We can prove this rigorously using Killing vectors, but we will ignore that. Note that this is the metric on the 2-sphere. So rotational invariance implies that our metric is of the form

$$\mathrm{d}s^2 = g_{tt}(t,r)\mathrm{d}t^2 + g_{rr}(t,r)\mathrm{d}r^2 + 2g_{tr}(t,r)\mathrm{d}r\mathrm{d}t + g_{\theta\theta}(t,r)\mathrm{d}\Omega_2^2.$$

So

$$g_{\mu\nu} = \begin{pmatrix} g_{tt} & g_{tr} & & \\ g_{rt} & g_{rr} & & \\ & & g_{\theta\theta} & \\ & & & g_{\theta\theta}\sin^2\theta \end{pmatrix}.$$

Invariance under time means we can get rid of all the time dependence in each term of g, so the metric is actually

$$\mathrm{d}s^2 = g_{tt}(r)\mathrm{d}t^2 + g_{rr}(r)\mathrm{d}r^2 + g_{\theta\theta}(r)\mathrm{d}\Omega_2^2.$$

Define a new coordinate $\widetilde{r}$ with $\widetilde{r}(r) = \sqrt{g_{\theta\theta}(r)}$. Write $g_{\mu\nu}(r(\widetilde{r})) = g_{\mu\nu}(\widetilde{r})$, which is an abuse of nootation. Then write

$$g_{\widetilde{r}\widetilde{r}}(\widetilde{r}) \coloneqq g_{rr}(r(\widetilde{r}))\left(\frac{\mathrm{d}r}{\mathrm{d}\widetilde{r}}\right)^2\mathrm{d}\widetilde{r}^2.$$

Then the metric is of the form

$$\mathrm{d}s^2 = -g_{tt}(\widetilde{r})\mathrm{d}t^2 + g_{\widetilde{r}\widetilde{r}}(\widetilde{r})\mathrm{d}\widetilde{r}^2 + \widetilde{r}^2\mathrm{d}\Omega^2.$$

This simplifies the equation a lot since we only have two metric components, so it becomes a system of two ordinary differential equations. We will plug this into the simple case

$$G_{\mu\nu} = 0.$$

Write r instead of $\widetilde{r}$ to simplify notation. It just turns out that Einstein's tensor is diagonal

$$G_{\mu\nu} = \begin{pmatrix} G_{tt} & & & \\ & G_{rr} & & \\ & & G_{\theta\theta} & \\ & & & G_{\phi\phi} \sin^2 \theta G_{\theta\theta} \end{pmatrix}.$$

This gives us the system

$$G_{tt} = G_{rr} = G_{\theta\theta} = 0.$$

This is solveable. For the first component, we have the form

$$G_{tt} = \frac{-g_{tt}(r)}{r^2 g_{rr}^2(r)} \left(r \frac{\mathrm{d}g_{rr}}{\mathrm{d}r} + g_{rr}^2(r) - g_{rr}(r) \right).$$

Setting this to zero and cancelling the prefactor yields the (nonlinear) differential equation

$$r \frac{\mathrm{d}g_{rr}}{\mathrm{d}r} = g_{rr}(r) - g_{rr}^2(r).$$

This can become the separable equation

$$\mathrm{d}g_{rr} \left(\frac{1}{g_{rr}(r)} - \frac{1}{1 - g_{rr}(r)} \right) = \frac{\mathrm{d}r}{r}.$$

Solving this yields

$$g_{rr}(r) = \frac{1}{1 + \frac{C}{r}}$$

for C an integration constant. Now to solve for $g_{tt}(r)$, we use

$$\mathrm{d}s^2 = -g_{tt}(r)\mathrm{d}t^2 + \left(\frac{1}{1 + \frac{C}{r}} \right) \mathrm{d}r^2 + r^2 \mathrm{d}\Omega^2.$$

We then get

$$r^2 G_{rr} = \frac{C}{C + r} + r \frac{g_{tt}'(r)}{g_{tt}(r)} = 0 \iff \mathrm{d}r \left(\frac{1}{C + r} - \frac{1}{r} \right) = \frac{\mathrm{d}g_{tt}}{g_{tt}}.$$

Solving yields

$$g_{tt}(r) = \frac{1}{B} \left(1 + \frac{C}{r} \right).$$

We can delete the $1/B$ by taking the time-invariant translation $t \to t\sqrt{B}$. Then

$$\mathrm{d}s^2 = -\left(1 + \frac{C}{r} \right) \mathrm{d}t^2 + \left(\frac{1}{1 + \frac{C}{r}} \right) \mathrm{d}r^2 + r^2 \mathrm{d}\Omega^2.$$

Now define $R_S \coloneqq -C$. Then

$$\mathrm{d}s^2 = -\left(1 - \frac{R_S}{r} \right) \mathrm{d}t^2 + \left(\frac{1}{1 - \frac{R_S}{r}} \right) \mathrm{d}r^2 + r^2 \mathrm{d}\Omega^2.$$

If we plug this into $G_{\theta\theta}$ then it becomes 0, so this is a valid solution. This is the *Schwartzschild metric*. Black holes arise when $R_S = r$: we get

$$\mathrm{d}s^2 = 0\mathrm{d}t^2 + \frac{1}{0}\mathrm{d}r^2 + r^2 \mathrm{d}\Omega^2.$$

So we see that as $r \to R_S$, an outside observer sees their clock slow down asymptotically.

The reason this is physical is because we can do a change of coordinates in r and make the prefactor for $\mathrm{d}r^2$ anything we want. So it is not clear whether or not this is just a bad coordinate system, or if it's truly a physical singularity.

How does the mass of the black hole come into play? Take $r \gg R_S$, and in the newtonian limit we have

$$g_{tt} = -\left(1 + 2\Phi\right).$$

We can take

$$2\Phi = -\frac{R_S}{r}.$$

In Newtonian gravity, if there is a massive souce of gravity, then

$$\Phi = \frac{G_N M}{r}.$$

So we end up with

$$R_S = 2G_N M.$$

Scalars are invariant under coordinate transformations. There is a nonzero scalar

$$R^{\mu\nu\sigma r} R_{\mu\nu\sigma\rho} = \frac{R R_S^2}{r^6},$$

which blows up at $r = 0$, *not* at $r = R_S$. This means that the event horizon is not a singularity, so if we fall across a horizon as an observer we actually will not notice anything different (assuming the universe is empty outside of the black hole, and neglecting spaggettificatoin). This means our infinity in the metric at $r = R_S$ is an artifact of the coordinates.

If we have a star, then outside we have $T_{\mu\nu}$, meaning the Schwartzschild solution actually applies. We can then find the horizon, and if the radius of the horizon is *inside* the sun, then that's fine since Schwartzschild doesn't work inside the sun since $T_{\mu\nu} \neq 0$ inside. Apparently we can compute

$$R_S = 2.6\text{km}.$$

The sun is slightly larger than that. Therefore, the sun is not a black hole.

Lecture 15: Blacker holes

Tue 04 Nov 2025 12:37

Can we find the geodesics of the Schwartzschild metric? These should correspond to stable orbits. Recalling

$$\mathrm{d}s^2 = -\left(1 - \frac{2G_M}{r}\right)\mathrm{d}t^2 + \frac{1}{1 - \frac{2G_M}{r}}\mathrm{d}r^2 + r^2\left(\mathrm{d}\theta^2 + \sin^2\mathrm{d}\phi^2\right),$$

write $f(r) = 1 - \frac{2G_M}{r}$, and use the action principle

$$L = \frac{1}{2}\int -g_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}\,\mathrm{d}\lambda = \frac{1}{2}\int f(r)(t')^2 - \frac{(r')^2}{f(r)} + r^2((\theta')^2 + \sin^2\theta(\phi')^2\,\mathrm{d}\lambda.$$

We can choose $\theta = \pi/2$ without loss of generality (orbit in xy-plane). I zoned out but then

$$\varepsilon = -g_{\mu\nu}P^\mu P^\nu, \qquad P^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}.$$

Then

$$\frac{\mathrm{d}\varepsilon}{\mathrm{d}\lambda} = P^\alpha \nabla_\alpha \left(-g_{\mu\nu} P^\mu P^\nu\right) = 0.$$

Then if $\varepsilon > 0$ we have massive particles, if $\varepsilon = 0$ we have massless particles, and for $\varepsilon < 0$ we have the power of love and friendship. Then you haver)

$$\frac{\mathrm{d}t}{\mathrm{d}\lambda} \equiv \frac{E}{f(r)}.$$

I dont know bbut somehow

$$\varepsilon = \frac{E^2 - (r')^2}{f'(r)} - \frac{\ell^2}{r^2}.$$

Then we get

$$\frac{1}{2}(r')^2 + V(r) = \frac{1}{2}E^2, \qquad V(r) = \frac{1}{2}f(r)\left(\frac{\ell^2}{r^2} + \varepsilon\right).$$

Idk but yeah. Check prof's notes.

Lecture 16: Einstein's equation in stars

Tue 18 Nov 2025 12:37

If we include quantum mechanics, we expect black holes to evaporate in time.

In black holes, $T_{\mu\nu} = 0$, but inside stars this is not true. We need to understand stars to even see if black holes can form. We can assume a rotationally invariant and time-independent metric

$$\mathrm{d}s^2 = -g_{tt}\mathrm{d}t^2 + g_{rr}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2$$

on the 2-sphere, where we have noted that there is no $\mathrm{d}t\mathrm{d}r$ coefficient since t and $-t$ should be symmetric. Additionally, we know that g_{tt}, g_{rr} depend only on r due to t- and rotational-invariance. Of course if $T_{\mu\nu} = 0$ then we get our Schwartzschild solution

$$g_{tt} = g_{rr} = 1 - \frac{2G_N M}{r}.$$

Inside a star this isn't true, so we will assume a star is a perfect fluid with $T^{\mu\nu}$ characterized by energy density $\rho(r)$, pressure $p(r)$, and 4-velocity $U^\mu(r)$. So somehow

$$T_{\mu\nu} = (p+\rho)U_\mu U_\nu + p g_{\mu\nu}.$$

This is easy to derive by passing to locally inertial coordinates with $T^\mu{}_\nu = \mathrm{diag}(\rho, p, p, p)$. Recall

$$U^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}.$$

The 3-velocity has to be invariant under rotations, and the only rotationally-invariant vector is 0, so

$$\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau} = \delta^\mu_0 \frac{\mathrm{d}(x^0 = t)}{\mathrm{d}\tau}.$$

Recall that $U^\mu U_\mu = -1$, so $U^0 U^0 g_{\mu\nu} = -1$. Therefore $U^0 U^0 g_{tt} = -1$, so

$$U^0 = -\sqrt{\frac{1}{-g_{tt}(r)}}, \qquad U_0 = \sqrt{-g_{tt}(r)}.$$

So we can obtain
$$T_{00} = (p+\rho)U_0^2 + \boldsymbol{\rho} g_{tt} = (p+\rho)g_{tt} + p(-g_{tt}) = \rho g_{tt}.$$
Additionally,
$$T_{11} = g_{rr}$$
since $U_1 U_1$ vanishes. This continues, so
$$T_{\mu\nu} = \begin{pmatrix} \rho(r)f(r) & & & \\ & p(r)/h(r) & & \\ & & p(r)r^2 & \\ & & & p(r)r^2\sin^2\theta \end{pmatrix}.$$
It is convenient to write $f(r) = -g_{tt}(r)$ and $h(r) = \frac{1}{g_{rr}(r)}$, such that $g_{tt}(r) = -f(r)$. We need to solve
$$G_{tt} = 8\pi G_N T_{00}, \qquad G_{rr} = 8\pi G_N T_{rr}, \qquad G_{\theta\theta} = 8\pi G_N T_{\theta\theta}.$$
It turns out that the hard part of this derivation is understanding ρ and p. This generally depends on the particular astrophysical body we focus on, but for now our equations are quite general. The first equation gives
$$G_{tt} = \left(1 - h(r) - rh'(r)\right) f(r)r^2 = 8\pi G_N \rho(r) f(r) \implies 1 - h(r) - rh'(r) = 8\pi G_N r^2 \rho(r).$$
It turns out the left hand side is $\frac{\mathrm{d}}{\mathrm{d}r}\left(r(1-h(r))\right)$, so we can solve
$$2(1-h(r)) = 2G_N 4\pi \int_0^2 r^2 \rho(r)\,\mathrm{d}r.$$
We write $m(r) = \frac{1-h(r)}{G_N}$, so
$$m(r) = 4\pi \int_0^r r^2 \rho(r)\,\mathrm{d}r.$$
Then
$$h(r) = 1 - \frac{2G_N m(r)}{r}.$$
This reminds us of $f(r)$ for the Schwartzschild metric, but now m is a function of r rather than a constant. Indeed, m is an integral over some sort of density function over radius, so this makes physical sense. By doing some investigation that is in the notes we get
$$p'(r) = -G_N(\rho(r)+p(r))\left(\frac{m(r) + 4\pi r^3 p(r)}{r(r - 2G_N m(r))}\right).$$
This is called the Tolman-Oppenheimer-Volkoff equation apparently. This is aids so instead there are other equations in the notes which are nicer.

Much of our study of p and ρ comes from the study of the equation of state $p(\rho)$ in thermodynamics.

Lecture 17: Perturbation theory for gravity

Thu 20 Nov 2025 12:30

We will apply perturbation theory to see gravitational waves arise from Einstein's equation. We will study them in the limit that $g_{\mu\nu}$ is very close to flat space plus a tiny correction $\eta_{\mu\nu} + h_{\mu\nu}$. This will be closely related to quantum gravity. For example, in electromagnetism, a photon is just a minimal quantum of an electromagnetic wave. Similarly, a graviton would be a minimal quantum of a gravitational wave.

Recall from electromagnetism

$$S_{\text{EM}} = \int -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \, \mathrm{d}^4 x, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Expanding this out yields

$$S_{\text{EM}} = \frac{1}{2} \int A_\mu \left(\eta^{\mu\nu} \partial^2 - \partial^\mu \partial^\nu \right) A_\nu \, \mathrm{d}^4 x.$$

Apparently you can do this in your head. Moreover, Lorentz invariance and Gauge invariance states that the action is fixed under transformations $A_\mu(x) \to A_\mu(x) + \partial_\mu \alpha(x)$ for any function α. Something happens but once it couples to charged matter the action obtains a term $A_\mu J^\mu$. Yea idk the full action is

$$S = \int -\frac{1}{4e^2} F_{\mu\nu} F^{\mu\nu} + A_\mu J^\mu \, \mathrm{d}^4 x.$$

Now recall the Einstein-Hilbert action

$$S_{\text{EH}} = \int \sqrt{-g} R \, \mathrm{d}^4 x.$$

We have to expand this out to second order Taylor expansion in h. At h^0 with $g_{\mu\nu} = \eta_{\mu\nu}$, we have $R = 0$ so $S_{\text{EH}} = 0$. For the linear term in h, recall

$$R_{\mu\nu\rho\sigma} = \frac{1}{2} \left(\partial_\rho \partial_\nu h_{\mu\sigma} + \partial_\sigma \partial_\mu h_{\nu\rho} - \partial_\sigma \partial_\nu h_{\mu\rho} - \partial_\rho \partial_\nu h_{\nu\sigma} \right) + O(h^2).$$

Contracting indicies,

$$R_{\mu\nu} = R_{\sigma\mu\sigma\nu} = \frac{1}{2} \left(\partial_\sigma \partial_\nu h^\sigma{}_\mu + \partial_\sigma \partial_\mu h^\sigma{}_\nu - \partial_\mu \partial_\nu h - \partial^2 h \right),$$

$$h = \operatorname{tr} h = \eta^{\mu\nu} h_{\mu\nu}.$$

Contracting again,

$$R = \partial_\mu \partial_\nu h^{\mu\nu} - \partial^2 h.$$

The action is then

$$S_{\text{EH}} = \int \partial_\mu \partial_\mu h^{\mu\nu} - \partial^2 h \, \mathrm{d}^4 x.$$

Integrating by parts, and ignoring boundary terms since we assume any further corrections fall off in the limit, the integral vanishes since it is a total derivative. This means we now pass to the h^2 term. We already know

$$\frac{1}{\sqrt{-g}} \frac{\delta S_{\text{EH}}}{\delta g^{\mu\nu}} = G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}.$$

We can expand out $R_{\mu\nu}$ as we did above. We then integrate to get the quadratic term in the action

$$S = \frac{1}{2}h^{\mu\nu}\frac{\delta S}{\delta h^{\mu\nu}} + O(h^3) = \frac{1}{2}\int \text{a total mess of an integrand}\, \mathrm{d}^4x.$$

This is nice, but it is cleaner to rederive it from Gauge invariance. For gravity, this is invariance under diffeomorphisms. Consider a coordinate transformation $(x')^\mu = x^\mu + \varepsilon^\mu(x)$ for some small ε. The metric transforms as

$$g'_{\mu\nu} = \frac{\mathrm{d}x^\alpha}{\mathrm{d}(x')^\mu}\frac{\mathrm{d}x^\beta}{\mathrm{d}(x')^\nu}g_{\alpha\beta} = (\delta^\mu_\alpha - \partial_\mu\varepsilon^\alpha)\left(\delta^\nu_\beta - \partial_\nu\varepsilon^\beta\right)g_{\alpha\beta}.$$

Now write $g = \eta + h$ and expand to the leading order, treating ε and h as the same size:

$$\eta_{\mu\nu} + h'_{\mu\nu} = \nu_{\mu\nu} + h_{\mu\nu} - \partial_\nu\varepsilon_\mu - \partial_\mu\varepsilon_\nu.$$

Thus h transforms as

$$h_{\mu\nu} \to h'_{\mu\nu} = h_{\mu\nu} - \partial_\mu\eta_\nu - \partial_\nu\eta_\mu.$$

We obtain the action from this by writing down a Lorentz-invariant action and then imposing this gauge invariance. I don't see a point in writing down the equations because we skip all the calculation anyways.

So $h_{\mu\nu}$ has 10 independent components

$$h_{00} \coloneqq \Phi, \qquad h_{0i} \coloneqq \omega_i, \qquad h^\alpha{}_k \coloneqq \Psi, \qquad h^i{}_k \coloneqq s^{ij}.$$

We are going to see that nearly all of these vanish in some way except the final one, which has $5 = 2(2) + 1$ components, and thus has spin 2.

Lecture 18: Gravitational waves

Tue 25 Nov 2025 12:33

Recall we are working in a metric $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ for h small, meaning we take $h^2 = 0$. Also typo above: $h_{00} = -2\Phi$.

Recall the gauge theory of E&M:

$$A_\mu \to A_\mu + \partial_\mu\lambda(x).$$

The gauge transformation for gravity is

$$h_{\mu\nu} \to h_{\mu\nu} - \partial_\mu\varepsilon_\nu - \partial_\nu\varepsilon_\mu.$$

Componentwise,

$$h_{00} \to h_{00} - 2\partial_0\varepsilon_0$$

$$\implies \Phi \to \Phi + \partial_0\varepsilon_0.$$

So we obtain

$$w_i \to w_i - \partial_0\varepsilon^i + \partial_i\varepsilon^0.$$

$$\Psi \to \Psi + \frac{1}{3}\partial_i\varepsilon^i.$$

$$s_{ij} \to \partial_i\varepsilon_j + \frac{1}{2}\partial_t\varepsilon^t\delta_{ij}.$$

Since e^μ has 4 components, there are 4 diffeomorphisms. Each gauge transformation removes a degree of freedom, so we lose 4 more degrees of freedom, leaving us with 2.

We can choose $\varepsilon^\mu(x)^2$ to make $G_{\mu\nu} = 0$. Gravitational waves are fluctuations in a vaccuum. We will impose $\partial_i s^{ij} = 0$ and $\partial_i w^i = 0$. This reduces our Einstein tensor

$$G_{00} = 2\nabla^2\Psi = 0, \qquad G_{0i} = -\frac{1}{2}\nabla^2 s_i + \partial_0\partial_i\Psi = 0.$$

From the first equation we obtain $\Psi = 0$, and from the second we obtain $\omega = 0$. Our equation is now much simpler:

$$G^i_i = 2\nabla^2\Phi$$
$$G^i_j = (\delta^i_j\nabla^2 - \partial_i\partial_j)\Phi + (\partial_0^2 - \nabla^2)s_{ij} = 0.$$

By definition s_{ij} is traceless, so

$$G^i_i = 2\nabla^2\Phi \implies \Phi = \frac{1}{\nabla^2}0 = 0.$$

Therefore $\left(\partial_0^2 - \nabla^2\right) s_{ij} = 0$. Recall that since $\Phi = \Psi = \omega^i = 0$,

$$h_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & s^{11} & s^{12} & s^{13} \\ 0 & s^{21} & s^{22} & s^{23} \\ 0 & s^{31} & s^{32} & s^{33} \end{pmatrix}.$$

This is easier to use if we pass to Fourier space.

$$\widetilde{s}_{ij} = \int e^{ik_\mu x^\mu} s_{ij}(x)\, \mathrm{d}^4x.$$

We can Fourier transform both sides.

$$\int e^{ik_\mu x^\mu} \left(\partial_0^2 - \nabla^2\right) s_{ij}\, \mathrm{d}^4x.$$

I will trust the professor that we obtain

$$k_0^2 - k^2 = 0.$$

Then

$$k_\mu k^\mu = 0.$$

This is like the 4-momentum, and in momentum,

$$p_0^2 - (\vec{p})^2 = m_{\text{rest}}^2.$$

Thus the graviton is massless. Since the only degree of freedom left over is s_{ij} with spin 2, gravitational waves are massless of spin 2. There is a fascinating implication we will not cover which is that if we assume there exists a massless spin 2 particle, then we obtain Einstein's entire theory of gravity.

$$g_{\mu\nu} = \begin{pmatrix} 1 & \\ & R \end{pmatrix}.$$